

Exploring the Different Roles of Particle Size in Photoelectrochemical and Photocatalytic Water Oxidation on BiVO₄

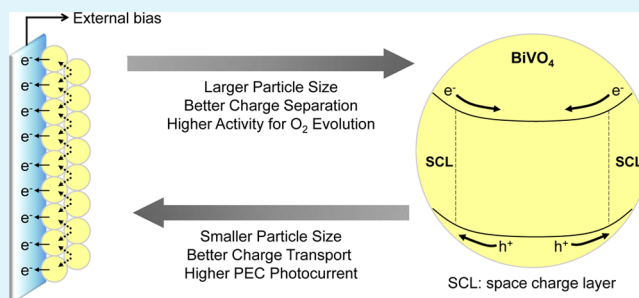
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S Supporting Information

ABSTRACT: Water oxidation on visible-light-active bismuth vanadate (BiVO₄) has commonly been demonstrated to be viable in powder suspension (PS) and particulate photoelectrochemical (PEC) systems. Here, we demonstrate that particle size reduction, which is known to be efficacious in promoting charge carrier extraction and boosting surface active sites, has an opposite effect on BiVO₄'s photoactivity in the two systems. With three BiVO₄ samples of distinctive particle sizes, smaller BiVO₄ particle size is shown to be beneficial for enhancing PEC photocurrent generation, but deleterious for photocatalytic O₂ evolution on suspended BiVO₄. Such contrary effect of particle size in the PEC and PS systems is revealed to be due to the different governing factors of the systems: charge transport in the former and charge separation in the latter. Smaller particle size was found to enrich the interparticle and the particle/FTO substrate contacts which improve charge transport and charge collection efficiency in BiVO₄ particulate electrode. On the contrary, larger particle size is necessary for improved photocatalytic O₂ evolution because of increased crystallinity and greater band bending, which are essential for charge separation.

KEYWORDS: bismuth vanadate, particle size, photocatalysis, photoelectrochemistry, water oxidation



INTRODUCTION

An ongoing effort in the research community has been devoted into the development of photocatalysts for solar water splitting to produce H₂ as a clean and renewable energy carrier. Because of the limited availability of photocatalyst materials capable of visible-light-driven water splitting into H₂ and O₂, great attention has thus been directed into developing visible-light-responsive photocatalysts active for evolution of either H₂ or O₂ from water in the presence of sacrificial reagents to realize the overall water splitting via a Z-scheme photocatalysis system. The said system involves a two-step photoexcitation of the H₂- and O₂-evolving photocatalysts present in a single medium containing a suitable electron mediator.^{1–4} Although H₂ is the primarily sought after reaction product, an efficient O₂-evolving photocatalyst is equally important to attain effective overall water splitting.

Monoclinic BiVO₄, with its valence band composed of hybridized Bi 6s and O 2p orbitals,^{5,6} has a relatively narrower bandgap (2.4–2.5 eV) compared to other metal oxide semiconductor photocatalysts and it is highly active for water oxidation. While early work on BiVO₄ mainly concentrated on the powder suspension (PS) system for photocatalytic O₂ evolution,^{5,7,8} more recent studies have also shown its feasibility for water oxidation via the photoelectrochemical (PEC) system, in which the n-type BiVO₄ is employed as the particulate photoanode.^{6,9–15} The performance of BiVO₄ is generally limited by its slow water oxidation kinetics¹⁶ and poor electron

transport properties.¹⁰ Despite successful demonstration of the introduction of various water oxidation catalysts (e.g., Co-Pi,^{16–19} NiOOH,¹² and FeOOH^{12,14,20}) and donor-type dopants (Mo^{21,22} and W^{11,18,23}) in circumventing the respective limitations, the efficiency of BiVO₄ is still greatly impaired by its intrinsically low carrier mobility.²⁴

In this regard, nanoscaling has been widely employed to reduce the photocatalyst size to shorten the charge transport pathways as well as to provide larger surface active sites, subsequently promoting minority carrier extraction and charge transfer kinetics.^{12,25–27} However, reduction of particle size is generally achieved at the expense of crystallinity, at which the density of surface defects is increased and results in augmented surface recombination. Such conflicting transformation of properties ensuing from nanoscaling thus renders particle size an intricate issue in governing the photoactivities in particulate PEC and PS systems.

The charge separation mechanisms in the two systems are essentially different. In PS system containing n-type semiconductor, charge separation relies on the thickness of space charge layer (SCL) formed at the photocatalyst/electrolyte interface to drive photogenerated holes to the electrolyte and electrons to the bulk.^{26,28} On the contrary, charge separation is

Received: July 22, 2016

Accepted: September 27, 2016

Published: September 27, 2016

facilitated by an externally applied potential in the particulate PEC system to draw electrons out from the illuminated photoanode to the counter electrode.²⁸ The difference in charge separation mechanisms of the two systems suggests that different particle qualities (for example, size and crystallinity) are required to improve their performance in the respective system. With these, Hong et al.²⁸ reported that particle size has an opposite effect on the water oxidation performance of WO₃ in the PS (proportional to the particle size) and particulate PEC (inversely proportional) systems.

To date, the beneficial effects of nanostructured BiVO₄ have been extensively shown in the particulate PEC system for water oxidation,^{9,12–15} whereas studies on the influence of particle size on photocatalytic O₂ generation performance of suspended BiVO₄ is relatively scarce. For example, nanosized BiVO₄ particles (<100 nm) prepared by Kho et al.²⁹ via flame spray pyrolysis showed poor photocatalytic O₂ evolution activity despite their high specific surface area. Similarly, Iwase and Kudo³⁰ observed poor photocatalytic performance of small and spherical BiVO₄ particles in comparison to the significantly larger and well-crystalline BiVO₄ particles. In this work, we provide an explicit study comparing the nanoscaling effect on the photoactivity of BiVO₄ in the particulate PEC and PS systems, which are seemingly similar for water oxidation yet different in the working principle and the charge separation mechanisms present within the photocatalyst. Contrary behavior of BiVO₄'s photoactivity dependence on the particle size was exhibited in the two systems (inversely and directly dependent on particle size in the PEC and PS systems, respectively). Detailed investigation on the interplay between the physical properties (size, crystallinity, interparticle, and BiVO₄/FTO substrate contacts) and the photoactivities of BiVO₄ in PEC and PS systems indicate different features govern the activity of BiVO₄ photocatalyst in the two systems, namely charge transport in the former and charge separation in the latter.

EXPERIMENTAL SECTION

Synthesis of BiVO₄. BiVO₄ was synthesized using solid–liquid state reaction in which the bismuth- and vanadium-containing precursors were mixed in an acidic aqueous solution (50 mL) at room temperature. To prepare sample A, equimolar amounts (5 mmol) of bismuth(III) oxide (Bi₂O₃) and vanadium(V) oxide (V₂O₅) were stirred in 0.5 M of nitric acid for 4 days. To synthesize sample B, 10 mmol of bismuth(III) nitrate pentahydrate (Bi(NO₃)₃·5H₂O) and 5 mmol of V₂O₅ were stirred in 0.75 M of nitric acid for 2 days. On the other hand, sample C was obtained by stirring equimolar amounts (5 mmol) of Bi₂O₃ and V₂O₅ in 1 M of acetic acid for 16 days. The resulting vivid yellow suspension from each of the preparation methods was collected by suction filtration, followed by washing with distilled water, and dried at 110 °C. Additionally, sample C was treated by air calcination at 400 °C for 1 h.

Materials Characterization. The morphology of the BiVO₄ samples was examined using a Hitachi S900 scanning electron microscope (SEM) operated at 4 kV. The resulting SEM images were then used for particle size measurements using Image-J software. Brunauer–Emmett–Teller (BET) surface area of each BiVO₄ sample was determined by a Micromeritics Tristar 3030 nitrogen adsorption apparatus at 77 K. The powder BiVO₄ sample was degassed in a Micrometric Smart Vacprep unit at 110 °C for 3 h before the surface area analysis. X-ray diffraction (XRD) measurements were obtained using a Phillips Xpert Multipurpose X-ray Diffraction System (MPD) at room temperature with Cu K α radiation (λ = 1.54 Å) at 45 kV and 40 mA. The crystallite size was estimated by Scherrer equation using the full width at half-maximum height of the peak at 2θ = 30.6°.

Diffuse reflectance spectra were analyzed using a UV–vis–NIR spectrophotometer (Shimadzu UV-3600) and were converted from reflectance to absorbance using the Kubelka–Munk method. X-ray photoelectron spectroscopy (XPS) was performed on a Thermo Scientific ESCALAB250Xi probe with monochromated Al K α radiation ($h\nu$ = 1486.68 eV). The binding energy was calibrated using the C 1s peak at 284.8 eV as reference. Cross sections of the BiVO₄/FTO with different particle sizes of BiVO₄ were observed using a FEI Nova NanoSEM 450 field-emission SEM, in which the electrode sample was tilted by 10°.

Preparation of Electrodes. The particulate electrodes used for PEC measurements, electrochemical impedance spectroscopy (EIS), and SEM imaging of the BiVO₄/FTO cross-sections were prepared by dropcasting the BiVO₄ powder samples on conductive fluorine-doped tin oxide (FTO) substrates. Typically, 6 mg of BiVO₄ was suspended in 6 mL of ethanol by sonication to obtain a well-dispersed suspension, which was then deposited onto 2 × 1.5 cm area of FTO substrate dropwise. For improving the adhesion of BiVO₄ particles onto the glass substrate, the obtained electrodes were heated at 300 °C in air for 2 h. The morphology and size of the BiVO₄ particles were preserved after heat treatment.

Photoelectrochemical Measurements (PEC System). PEC measurements were performed in a standard three-electrode cell with the BiVO₄ electrode as the working electrode, a platinum foil as the counter electrode and a saturated Ag/AgCl electrode as the reference electrode. N₂-saturated 0.1 M sodium sulfate (Na₂SO₄) with or without methanol (10 vol %) was used as the electrolyte for all the electrochemical measurements. The working electrode was illuminated from the FTO side (backside illumination) with visible-light from a 300 W Xe lamp (Oriel) coupled with a 420 nm cutoff filter. The current–potential (I–V) curves were recorded using an Autolab potentiostat (Model PGSTAT302N) at a potential scan rate of 0.02 V s^{–1}. EIS Nyquist plots were acquired at 0 V vs Ag/AgCl with AC amplitude of 5 mV in the frequency range of 100 kHz to 0.01 Hz using a frequency response analyzer (FRA).

Photocatalytic Reaction (PS System). Photocatalytic O₂ evolution was carried out in an enclosed top-irradiation cell with a Pyrex window. 100 mg of BiVO₄ powder was dispersed in 100 mL of aqueous solution containing 0.05 M AgNO₃ and purged with argon gas for 30 min before light illumination. The suspension was irradiated under continuous stirring using a 300 W xenon lamp (Oriel) coupled with a 420 nm cutoff filter. The amount of evolved O₂ was analyzed by gas chromatography (TCD, Shimadzu GC-8A, Ar carrier gas).

RESULTS AND DISCUSSION

BiVO₄ samples with different particle sizes were prepared via a similar solid–liquid state reaction at room temperature, at which bismuth- and vanadium-containing precursors were dissolved and precipitated in acidic aqueous solution to form BiVO₄. Stirring of bismuth(III) oxide (Bi₂O₃) and vanadium(V) oxide (V₂O₅) in nitric acid (HNO₃) resulted in large and square platelike BiVO₄ particles with smooth and clear faces,³¹ denoted as sample A shown in Figure 1a. Although replacement of Bi₂O₃ with bismuth(III) nitrate pentahydrate (Bi(NO₃)₃·5H₂O) as the Bi precursor led to the formation of smaller

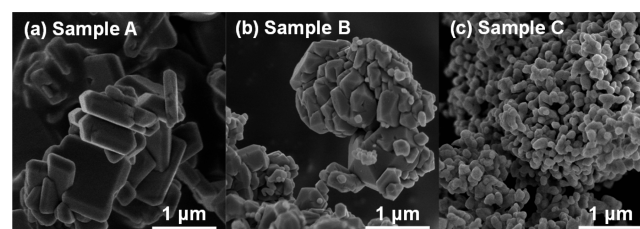


Figure 1. SEM micrographs of (a) sample A, (b) sample B, and (c) sample C.

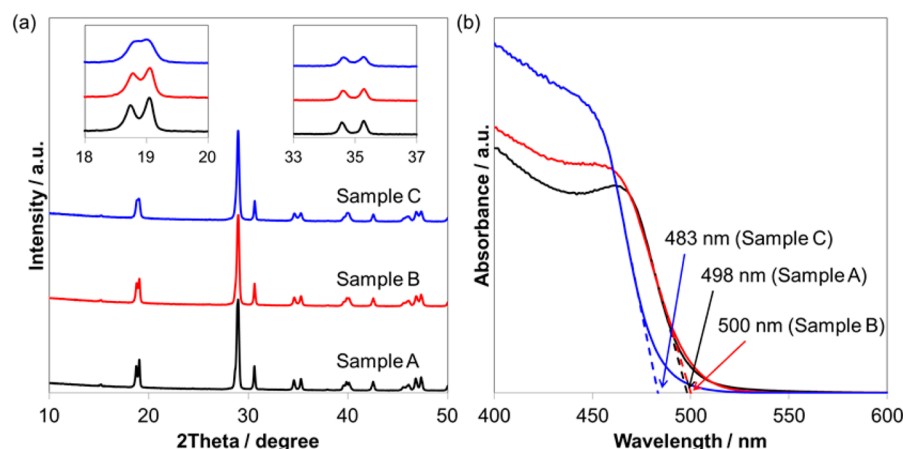


Figure 2. (a) XRD diffraction patterns and (b) UV-vis diffuse reflectance spectra of samples A (black line), B (red line), and C (blue line). Insets in panel a illustrate the enlarged XRD diffraction patterns showing the peak splitting at around 18.5° (left) and 35° (right).

Table 1. Physical and Optical Properties of the As-Prepared BiVO₄ Samples

	Bi precursor	acid aqueous solution, concentration (M)	particle size (nm)	BET surface area (m ² g ⁻¹)	crystallite size (nm)	band gap (eV)
A	Bi ₂ O ₃	HNO ₃ , 0.5	833 ± 249	2.1	91	2.49
B	Bi(NO ₃) ₃ ·5H ₂ O	HNO ₃ , 0.75	374 ± 171	2.9	89	2.48
C	Bi ₂ O ₃	CH ₃ COOH, 1.0	123 ± 24	5.4	72	2.57

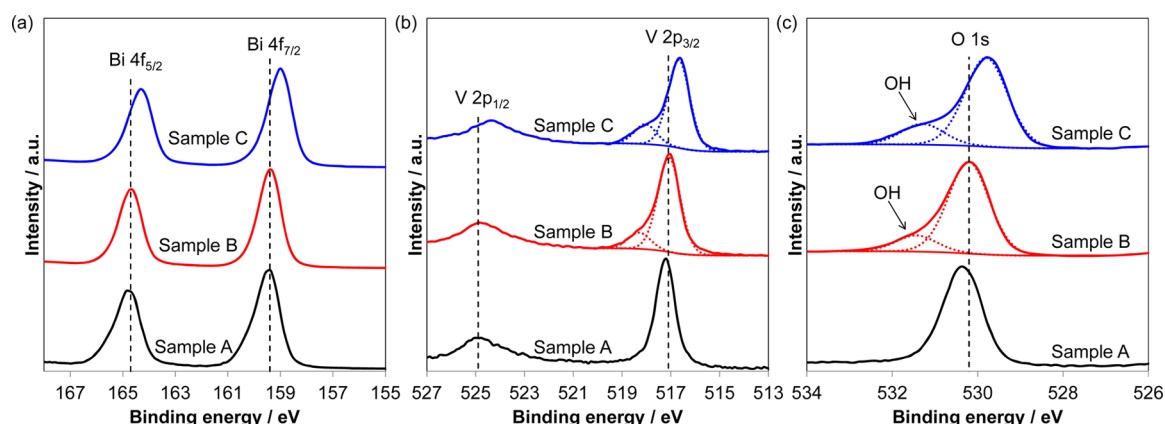


Figure 3. XPS spectra of the (a) Bi 4f, (b) V 2p, and (c) O 1s core levels of samples A, B, and C.

BiVO₄ particles with irregular shape (sample B; Figure 1b),³² substitution of HNO₃ with acetic acid (CH₃COOH) as the reaction medium produced the smallest spherical BiVO₄ particles (sample C; Figure 1c).³⁰ To achieve the same crystal phase as that of sample A and B, we treated sample C with postcalcination at an elevated temperature of 400 °C for its phase transition from scheelite-tetragonal to scheelite-monoclinic. Size measurement of 200 particles randomly selected from the scanning electron microscopy (SEM) micrographs indicate that samples A, B, and C have an average size of 833 ± 249 nm (determined by the width of the square particles), 374 ± 171 nm and 123 ± 24 nm, respectively. The histograms corresponding to the particle size distribution of each sample are included in Figure S1. Consistently, BET surface area analysis shows an increasing trend from 2.1 m² g⁻¹ for sample A, to 2.9 m² g⁻¹ for sample B and 5.4 m² g⁻¹ for sample C, corroborating the decreasing particle size from sample A to sample C.

Figure 2a displays the XRD patterns of the BiVO₄ samples. The discernible peak splitting at 18.5, 35, and 46° of 2θ signifies

that all samples have a single scheelite-monoclinic phase.⁸ Despite the supposedly split peak at 18.5° as a characteristic diffraction fingerprint of scheelite-monoclinic BiVO₄, the seemingly singlet for sample C can be attributed to the low crystallinity of the BiVO₄ particles obtained using aqueous CH₃COOH solution.³⁰ Note that the peak resolution worsens from sample A to sample C, which is readily demonstrated by the split peaks at 18.5 and 35° (illustrated in the insets), indicating that the crystallinity deteriorates as the particle size decreases. This observed particle size dependent crystallinity is supported by the gradual decrease of crystallite size from sample A to sample C (Table 1), as estimated using the Scherrer equation. The considerably larger particle size of each sample in comparison to the crystallite size suggests that the BiVO₄ samples used in this study are composed of multidomain crystals (i.e., nonsingle crystal). Although all BiVO₄ samples exhibit the typical optical absorption in the visible-light region, the absorption edge of sample C is slightly blue-shifted compared to that of sample A and B (Figure 2b). Such blue shift in absorption edge of sample C is in agreement with that

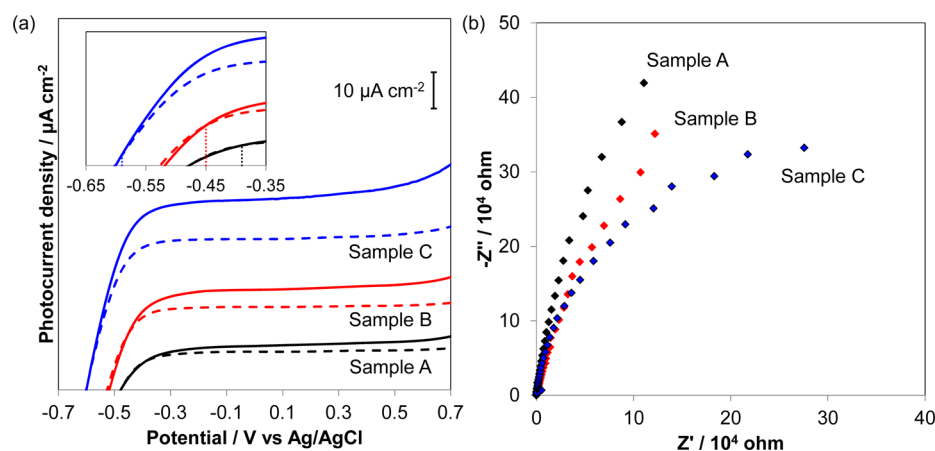


Figure 4. (a) I – V curves under dark (dashed line) and light (full line) conditions with a scan rate of 0.02 V s^{-1} and (b) Nyquist plots of samples A, B and C obtained in $0.1 \text{ M Na}_2\text{SO}_4$ solution. The inset in a illustrates the magnified view of the I – V curves showing the onset potential of each sample.

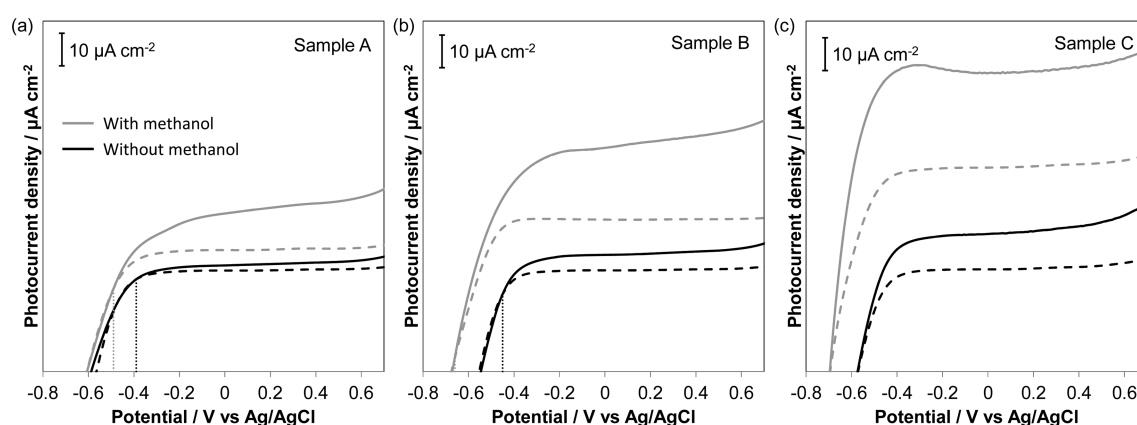


Figure 5. Comparison of the I – V curves of (a) sample A, (b) sample B, and (c) sample C under dark (dashed line) and light (full line) conditions with a scan rate of 0.02 V s^{-1} in $0.1 \text{ M Na}_2\text{SO}_4$ without additives (black lines) and after addition of 10 vol % methanol (gray lines).

reported by Iwase and Kudo.³⁰ The band gap energies were estimated as 2.49, 2.48, and 2.57 eV for samples A, B, and C, respectively. Table 1 summarizes the physical and optical properties of the as-prepared BiVO_4 for ease of comparison.

X-ray photoelectron spectroscopy (XPS) measurements were carried out to compare the surface qualities of the BiVO_4 samples with different particle sizes. Figure 3 shows the core levels Bi 4f, V 2p, and O 1s XPS spectra of all samples. The Bi 4f_{7/2}, Bi 4f_{5/2}, V 2p_{3/2}, V 2p_{1/2}, and O 1s peaks for both samples A and B are found at 159.4, 164.7, 517.1, 524.9, and 530.2 eV, respectively, which correspond to the Bi³⁺, V⁵⁺, and O²⁻ in BiVO_4 .^{33,34} In comparison to samples A and B, these peaks of sample C are consistently shifted by -0.4 eV . This can be ascribed to the surface effect of nanosized materials, whereby an increase in the ratio of surface to bulk atoms has been found to result in negative binding energy shift.³⁵ Deconvolution of the V 2p_{3/2} peak (Figure 3b) suggests that there is an additional peak for both samples B and C at higher binding energies located at 518.3 eV for sample B and 518.0 eV for sample C. These peaks are attributable to the presence of V⁵⁺ in V_2O_5 originated from the V precursor.³³ Likewise, deconvolution of the asymmetric O 1s peaks of both samples B and C (Figure 3c) reveals the presence of an additional peak located at the binding energies 1.2 and 1.5 eV higher than the O 1s peaks of the respective samples. The chemical shift within the range of 1.2–1.5 eV agrees well with that between hydroxyl oxygen (OH) and lattice oxygen (O²⁻) of metal oxide,³⁶ suggesting

that the additional peak for both samples B and C corresponds to the OH groups present at the defect sites.³⁷ It is noteworthy that sample C shows a relatively higher OH/O²⁻ ratio compared to sample B (0.23 and 0.18, respectively), indicating that the former has a higher density of surface defects ensuing from particle size reduction. These XPS results thus suggest the increase of surface defect density from sample A to C, which explains their decreasing crystallinity as observed from the XRD analysis (Figure 2a).

To evaluate the particle size effect on the PEC water oxidation performance of BiVO_4 under visible-light illumination (of wavelength, $\lambda > 420 \text{ nm}$), the powder samples were each made into an electrode via a dropcast method using conductive fluorine-doped tin oxide (FTO) glass as the substrate. Figure 4a depicts the current–potential (I – V) curves of the BiVO_4 particulate photoanodes under dark and light conditions. Upon light illumination, all samples exhibit anodic photocurrent which is a typical photoresponse of the n-type BiVO_4 . Improved magnitude of photocurrent density was observed as the BiVO_4 particle size decreased over the range of potentials studied. As opposed to sample A that portrays steady growth of photocurrent with the applied potential, samples B and C reach a saturated photocurrent at a low potential of -0.30 V vs Ag/AgCl, suggesting a more efficient charge separation and transportation in the photoanodes made with smaller BiVO_4 particles.³⁸ Additionally, the onset potential (estimated by the interception between dark and light curves³⁹) was found to be

progressively shifted to a more negative potential with the decrease of BiVO_4 particle size, as evidenced by the magnified view of I – V curves in the range of -0.65 to -0.35 V vs Ag/AgCl (inset in Figure 4a). A low onset potential is favorable for efficient PEC water splitting reactions in view of the requirement of low energy input (i.e., the applied potential).^{38,39} The negative shift of both the saturation and onset potentials for the photoanodes with smaller BiVO_4 particles thus infers enhanced PEC performance through decreasing BiVO_4 particle size. In order to understand the origin of the improved PEC performance for BiVO_4 with smaller particle size, the electrochemical properties of each BiVO_4 photoanode was determined using electrochemical impedance spectroscopy (EIS). Figure 4b presents the Nyquist plots of all samples measured at 0 V vs Ag/AgCl under dark condition. The arc radius decreases from sample A to sample C, suggesting an improving charge transfer ability of the photoanode accompanying the BiVO_4 particle size decrease, which correlates well with the activity trend in the PEC photocurrent generation.

The charge transfer efficiency of the photoanode is determined by both the electrons and holes, whereby the former migrate to the BiVO_4 /FTO interface to be drawn off as current and the latter to the BiVO_4 /electrolyte interface to conduct water oxidation reaction. The possibility of hole transfer as the limiting factor of the charge transfer efficiency of a photoanode can be eradicated by the addition of a hole scavenger. Figure 5 compares the I – V curves of all BiVO_4 photoanodes obtained with and without methanol (hole scavenger) addition. Consistent with that reported in the literature,^{6,40} significant improvement of photocurrent and negative shift of onset potential were observed for each sample in the presence of the hole-scavenging methanol as compared to that measured in the absence of methanol. Importantly, in the presence of methanol, BiVO_4 with smaller particle size also exhibits higher photocurrent density as well as lower saturation and onset potentials, similar to that observed in the absence of methanol. The results thus infer electron transfer as the governing factor of the improved charge transfer ability of BiVO_4 photoanode with smaller particle size.

Given that the photogenerated electrons have to be transported to the back conducting substrate at the photoanode to be drawn off as a current, the nature of the interparticle contacts as well as the contact between the particles and the substrate are the key features controlling the electron transport properties of the photoanode and thus the electron transfer efficiency at the particle/substrate interface.³⁰ With respect to these aspects, the cross sections of sample A, B, and C on FTO substrates were observed under SEM and the corresponding images are displayed in Figure 6. At a lower magnification, void spaces can be observed within the network of sample A particles (Figure 6a). The size of these void spaces shrinks as the BiVO_4 particle size decreases, as depicted by the sample B electrode in Figure 6b and sample C electrode in Figure 6c, suggesting a more compact particle network and better interparticle contacts with the presence of smaller particles. At a higher magnification focusing on the BiVO_4 /FTO interface, the contact of sample A particles and the FTO substrate (Figure 6d) is evidently poor because of stacking up of the large particles. Such BiVO_4 /FTO contact is improved in sample B and C electrodes (Figure 6e, f, respectively). Because of the comparable size of sample C particles with the rough surface of the FTO substrate, elemental energy-dispersive X-ray spectroscopy (EDS) mapping (Figure S2) was carried out to

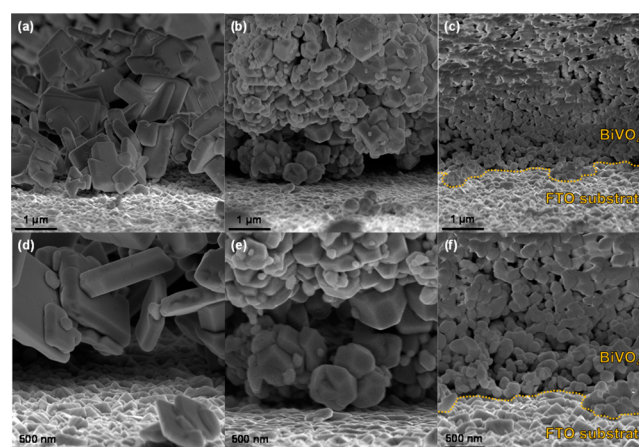


Figure 6. SEM micrographs of the cross sections of (a) sample A, (b) sample B, and (c) sample C BiVO_4 particles on FTO conducting glass. (d–f) Higher magnification of the SEM micrographs of sample A, B, and C interfaces with FTO, respectively.

aid in identifying the interface between sample C particles with the FTO substrate, as outlined in Figure 6f. Clearly, sample C particles with the smallest size are uniformly distributed on the FTO substrate, establishing the most intimate contact with the substrate compared to the other two samples. The SEM images of the BiVO_4 /FTO interfaces provide compelling evidence that the superior PEC performance of BiVO_4 with smaller particle sizes arises from the improved BiVO_4 particles contact with the electron collecting FTO substrate on top of the enriched interparticle contacts. These contacts ensure effective interparticle charge transport and thus efficient collection of photogenerated electrons at the FTO back substrate to render higher photocurrent generation.

To delve into the reactivity of the three BiVO_4 samples with different particle sizes in PS system, we carried out a separate activity test for water oxidation from aqueous silver nitrate (AgNO_3) solution under visible light (of wavelength, $\lambda > 420$ nm) illumination, in which Ag^+ ions were the electron scavenger. Figure 7 illustrates the amount of O_2 generated as a function of time for each sample. Sample A unambiguously exhibits the highest activity with the fastest O_2 evolution rate

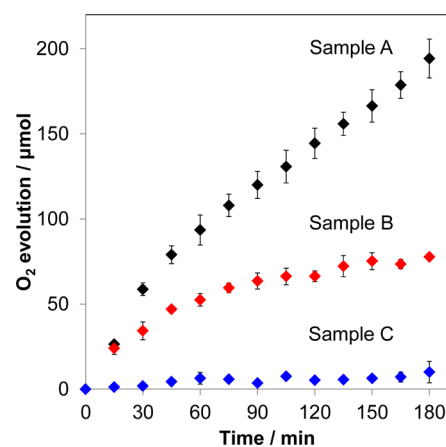
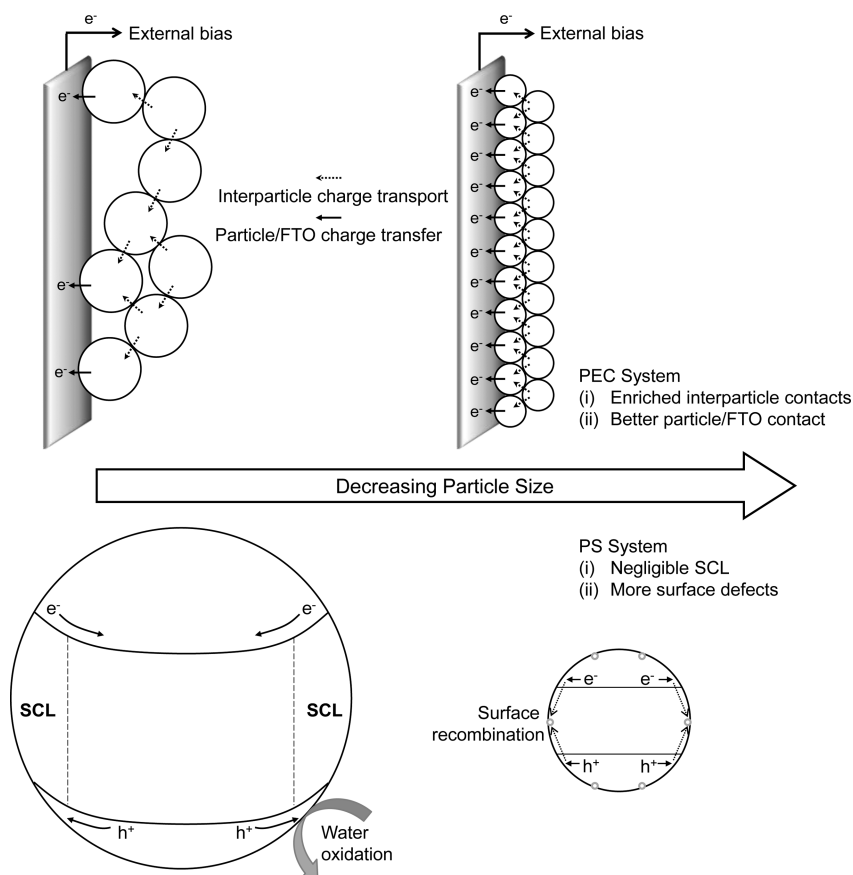


Figure 7. Time profiles of O_2 evolution from aqueous AgNO_3 solution for samples A, B, and C, under visible light (of wavelength, $\lambda > 420$ nm) illumination. Every data point was an average of 3 experimental runs with the error bar corresponding to standard deviation.

Scheme 1. Different Roles of Particle Size in Governing the Photoactivity of BiVO_4 in PEC and PS Systems, At Which the Former Is Predominated by Charge Transport Properties and the Latter Is Dominated by Charge Separation Efficiency^a



^aSCL denotes the space charge layer.

followed by samples B and C. Substantial amount of O_2 was generated by samples A and B (194 μmol and 76 μmol , respectively, over 3 h of illumination) except for sample C with an amount of merely 10 μmol , indicating larger BiVO_4 particles are beneficial for PS water oxidation activity. This observed trend of O_2 evolution activities of the suspended BiVO_4 particles (sample C < sample B < sample A) is in stark contrast to their tendency in the PEC system (sample A < sample B < sample C).

The opposing behaviors of PEC and PS photoactivity dependence on the particle size of BiVO_4 indicate different governing factors in the two systems, as schematically depicted in Scheme 1. Despite having more surface recombination centers arising from the higher density of surface defect sites, BiVO_4 particulate photoanode with smaller particles was shown to have better PEC performance. This is attributed to the presence of an externally applied potential to facilitate charge separation in the PEC system, in which the performance of the BiVO_4 photoanode is predominated by its charge transport properties. Smaller BiVO_4 particles immobilized on FTO substrate were found to have shorter interparticle distance (i.e., enriched interparticle contacts) and better particle/FTO contact, allowing efficient interparticle charge transport and electron collection efficiency at the substrate to result in an enhanced PEC photocurrent. In contrast, the increased photocatalytic O_2 evolution activity for suspended BiVO_4 with larger particle size demonstrates charge separation as the key issue in the PS system rather than charge transport. The

charge separation of a suspended photocatalyst is mainly dependent on the SCL formation. A thicker SCL is typically developed in larger particles to induce greater band bending beneficial for electron–hole pairs separation.⁴¹ In nanoparticle where its size is smaller than the SCL thickness, the resulting SCL and band bending are negligible. In fact, significantly higher photocatalytic water oxidation activity of large rutile TiO_2 particles in comparison to that of small anatase particles has been attributed by Ohno et al. to the presence of band bending in the large particles.^{42,43} Additionally, the photo-generated electrons and holes have to be present on the surface of the same particle to conduct redox reactions on the suspended photocatalyst in the PS system. These electron–hole pairs are prone to recombination at the surface defects, which was shown to be particularly rich on the surfaces of small BiVO_4 particles with low crystallinity. The presence of band bending and fewer surface defects in large particle thus signifies enhanced charge separation to improve photocatalytic O_2 evolution.

CONCLUSIONS

Nanoscaling was demonstrated to render opposite effects in the water photooxidation activity of BiVO_4 particles via PEC and PS systems. Specifically, improved PEC photocurrent generation was observed as the size of the BiVO_4 particles immobilized on the photoanode decreased. SEM images illustrated enriched interparticle contacts and better contact between the particles and the FTO substrate in the presence of

BiVO₄ with smaller particle size, resulting in better charge transport and charge collection efficiency that were supported by the EIS results. On the other hand, the photocatalytic O₂ evolution activity of the suspended BiVO₄ particles deteriorated with decreased particle size because of the loss of crystallinity and band bending necessary for charge separation. This work demonstrates that the water oxidation efficiency in the PEC and PS systems relies on different key issues, namely charge transport and charge separation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.6b09076.

Particle size distribution histograms and EDS mapping of BiVO₄/FTO interface (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the Australian Research Council for financial support (DP110101638). We acknowledge the UNSW Mark Wainwright Analytical Centre for providing access to all the analytical instruments.

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